

ISLAMIC UNIVERSITY OF TECHNOLOGY



OPERATING SYSTEM LAB
CSE 4502

Lab 2 Assignment

Submitted by:

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1. List of Commands

Part 1: Basic Commands (6 Commands)

1. **mkdir** — Create the workspace.
2. **cd** — Navigate into a directory.
3. **man** — Display the manual page for a command.
4. **history** — View previously executed commands.
5. **stat** — Display detailed file status and metadata.
6. **uname** — Print system information.

Part 2: Text Processing and Filters (8 Commands)

7. **wc** — Word count (lines, words, bytes).
8. **sort** — Sort lines of text files.
9. **uniq** — Remove duplicate lines.
10. **diff** — Find differences between two files.
11. **cut** — Extract specific columns from lines.
12. **tr** — Translate or delete characters.
13. **sed** — Stream editor for substitutions.
14. **more** — View file content page-by-page.

Part 3: Archiving, Compression, and Linking (6 Commands)

15. **tar -czf** — Create a compressed archive.
16. **tar -xf** — Extract files from archive.
17. **gzip** — Compress a file using GZIP.

- 18. **gunzip** — Decompress a .gz file.
- 19. **ln -s** — Create a symbolic link.
- 20. **ln** — Create a hard link.

Part 4: System Status and Process Control (6 Commands)

- 21. **df** — Report disk space usage.
- 22. **free** — Show free and used memory.
- 23. **top** — Display running processes.
- 24. **ps** — Snapshot of current processes.
- 25. **kill** — Terminate a process by PID.
- 26. **ip addr** — Show network interface information.

Part 5: I/O Control, Aliases, and Utilities (4 Commands)

- 27. **alias** / **unalias** — Create and remove command shortcuts.
- 28. **&&** — Append output to a file.
- 29. **—** — Pipe operator (output to input).
- 30. **date** / **cal** — Show system time/date and calendar.

2. Exact Command Syntax

```
1 mkdir -p lab2/text_data
2 cd lab2
3 man tar
4 history
5 stat text_data
6 uname -a
7 cat fileB.txt fileA.txt | wc
8 cat fileB.txt fileA.txt | sort
9 cat fileA.txt fileB.txt | sort | uniq
10 diff fileA.txt fileB.txt
11 cut -d " " -f 1 combined.txt
12 cat combined.txt | tr 'a-z' 'A-Z'
13 sed 's/a/x/g' fileA.txt
14 more combined.txt
15 tar -czf text_data.tar.gz text_data/
16 tar -xf text_data.tar.gz -C extracted_data/
17 gzip fileA.txt
18 gunzip fileA.txt.gz
19 ln -s fileA.txt link_soft
20 ln fileB.txt link_hard
21 df -h
22 free -h
23 top
24 ps aux
25 kill 560
26 ip addr
27 alias ll='ls -lh'
28 unalias ll
29 echo "emnei" >> fileA.txt
30 ps aux | grep sleep
31 date
32 cal
```

3. Execution Proof

```
rahatut@rahatut-laptop:~$ mkdir -p lab2/text_data
rahatut@rahatut-laptop:~$ cd lab2
rahatut@rahatut-laptop:~/lab2$ man tar
hrahatut@rahatut-laptop:~/lab2$ history
 1  chmod +x localjudge.sh
 2  cd /mnt/c/Users/User/iut-algorithms-lab-1a-summer-23-24
 3  chmod +x localjudge.sh
 4  ./localjudge.sh -path "./Lab 3/Task A"
 5  mkdir lab2
 6  cd lab2
 7  mkdir text_data
 8  man tar
 9  history
10  history | tail
11  stat text_data
12  uname -a
13  echo -e "a\nb" > fileA.txt
14  echo -e "c\nd" > fileB.txt
15  echo -e "a\nc" > fileB.txt
16  cat fileB.txt
17  cat fileA.txt fileB.txt
18  cat fileB.txt fileA.txt | sort
19  cat fileB.txt fileA.txt | wc
20  cat fileA.txt fileB.txt | uniq
21  cat fileA.txt fileB.txt | sort | uniq
22  diff fileA.txt fileB.txt
23  cut -d " " -f 1 fileA.txt
24  cut -d " " -f 2 fileA.txt
25  echo -e "a 1\nb 2\nc 3" > fileA.txt
26  echo -e "a 10\nb 20\nc 30" > fileB.txt
27  cut -d " " -f 2 fileA.txt
28  cut -d " " -f 1 fileA.txt
29  cut -d " " -f 2 fileB.txt
30  cat fileA.txt | tr 'a-z' 'A-Z'
31  cat fileA.txt | tr 'A-B' 'a-b'
32  cat fileA.txt
33  cat fileA.txt | tr 'a-a' 'A-A'
34  sed s/b/z/g fileA.txt
35  sed s/z/a/ fileA.txt
36  sed s/b/a/g fileA.txt [A
37  cat fileA.txt
38  more fileA.txt
39  tar -czf text_data.tar.gz text_data/
40  ls
41  mkdir extracted_data
42  tar -xf text_data.tar.gz -C extracted_data/
43  ls extracted_data/text_data
44  ls
45  cd extracted_data
46  ls
47  cd text_data
48  ls
49  cd .
50  cd -
51  cd ..
52  gzip fileA.txt
53  ls
54  gunzip fileA.txt.gz
55  ls
56  ln -s fileA.txt link_soft
57  ls -l
58  ln fileB.txt link_hard
```

Figure 1: Execution Screenshot 1

```

58 ln fileB.txt link_hard
59 ls
60 df
61 df -h
62 free -h
63 top
64 ps
65 ps aux
66 kill 1123
67 sleep 100
68 sleep 100 &
69 ps aux | grep sleep
70 kill 1163
71 kill 1161
72 ps aux | grep sleep
73 ip addr
74 history | grep sleep
75 echo "tessting" >> fileA.txt
76 cat fileA.txt
77 alias ll='ls -lh'
78 ll
79 unalias ll
80 ll
81 date
82 cal
83 clear
84 PS1="myname@MYPC:\w$ "
85 cd ..
86 PS1="myname@MYPC:\w$ "
87 PS1="rahatut@rahatut-laptop:\w$ "
88 mkdir -p lab2/text_data
89 cd lab2
90 man tar
91 history
rahatut@rahatut-laptop:~/lab2$ stat text_data
  File: text_data
  Size: 4096          Blocks: 8          IO Block: 4096   directory
Device: 8,64    Inode: 46409          Links: 2
Access: (0755/drwxr-xr-x)  Uid: ( 1000/  rahatut)   Gid: ( 1000/  rahatut)
Access: 2025-11-28 12:58:20.850701596 +0600
Modify: 2025-11-28 12:11:09.111117375 +0600
Change: 2025-11-28 12:11:09.111117375 +0600
 Birth: 2025-11-28 12:11:09.111117375 +0600
rahatut@rahatut-laptop:~/lab2$ uname -a
Linux DESKTOP-5903S8A 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu J
rahatut@rahatut-laptop:~/lab2$ echo -e "apple 10\nbanana 20\ngrape 30" > fileA.txt
rahatut@rahatut-laptop:~/lab2$ echo -e "apple 15\norange 25\nbanana 20" > fileB.txt
rahatut@rahatut-laptop:~/lab2$ cat fileA.txt fileB.txt > combined.txt
rahatut@rahatut-laptop:~/lab2$ sort combined.txt > sorted.txt
uniq sorted.txt > unique.txt
rahatut@rahatut-laptop:~/lab2$ cat combined.txt
apple 10
banana 20
grape 30
apple 15
orange 25
banana 20
rahatut@rahatut-laptop:~/lab2$ sort combined.txt
apple 10
apple 15
banana 20
banana 20
grape 30
orange 25

```

Figure 2: Execution Screenshot 2

```

rahatut@rahatut-laptop:~/lab2$ uniq sorted.txt > unique.txt
rahatut@rahatut-laptop:~/lab2$ cat unique.txt
apple 10
apple 15
banana 20
grape 30
orange 25
rahatut@rahatut-laptop:~/lab2$ diff fileA.txt fileB.txt
1c1,2
< apple 10
---
> apple 15
> orange 25
3d3
< grape 30
rahatut@rahatut-laptop:~/lab2$ cut -d " " -f 1 combined.txt
apple
banana
grape
apple
orange
banana
rahatut@rahatut-laptop:~/lab2$ cat combined.txt | tr 'a-z' 'A-Z'
APPLE 10
BANANA 20
GRAPE 30
APPLE 15
ORANGE 25
BANANA 20
rahatut@rahatut-laptop:~/lab2$
rahatut@rahatut-laptop:~/lab2$ sed 's/a/x/g' fileA.txt
xpple 10
bxnxnx 20
grxpe 30
rahatut@rahatut-laptop:~/lab2$ more combined.txt
apple 10
banana 20
grape 30
apple 15
orange 25
banana 20
rahatut@rahatut-laptop:~/lab2$ tar -czf text_data.tar.gz text_data/
rahatut@rahatut-laptop:~/lab2$ mkdir -p extracted_data
rahatut@rahatut-laptop:~/lab2$ tar -xf text_data.tar.gz -C extracted_data/
rahatut@rahatut-laptop:~/lab2$ ls
combined.txt  fileA.txt  link_hard  sorted.txt  text_data.tar.gz
extracted_data  fileB.txt  link_soft  text_data  unique.txt
rahatut@rahatut-laptop:~/lab2$ gzip fileA.txt
rahatut@rahatut-laptop:~/lab2$ gunzip fileA.txt.gz
rahatut@rahatut-laptop:~/lab2$ ln -s fileA.txt link_soft
ln: failed to create symbolic link 'link_soft': File exists
rahatut@rahatut-laptop:~/lab2$ ln fileB.txt link_hard
ln: failed to create hard link 'link_hard': File exists
rahatut@rahatut-laptop:~/lab2$ rm link_soft link_hard
rahatut@rahatut-laptop:~/lab2$ ln -s fileA.txt link_soft
rahatut@rahatut-laptop:~/lab2$ ln fileB.txt link_hard
rahatut@rahatut-laptop:~/lab2$ ls
combined.txt  fileA.txt  link_hard  sorted.txt  text_data.tar.gz
extracted_data  fileB.txt  link_soft  text_data  unique.txt
rahatut@rahatut-laptop:~/lab2$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.86  0    3.86  0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            3.86  4.0K  3.86  1% /mnt/wsl

```

Figure 3: Execution Screenshot 3

```

none          3.86      0  3.86    0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none          3.86    4.0K  3.86    1% /mnt/wsl
drivers       2386   1806    586   76% /usr/lib/wsl/drivers
/dev/sde      10076   1.36  9556    1% /
none          3.86    76K  3.86    1% /mnt/wslg
none          3.86      0  3.86    0% /usr/lib/wsl/lib
rootfs        3.86   2.7M  3.86    1% /init
none          3.86   544K  3.86    1% /run
none          3.86      0  3.86    0% /run/lock
none          3.86      0  3.86    0% /run/shm
none          3.86    96K  3.86    1% /mnt/wslg/versions.txt
none          3.86    96K  3.86    1% /mnt/wslg/doc
C:\           2386   1806    586   76% /mnt/c
D:\           2356   101M  2356    1% /mnt/d
E:\           2426   923M  2426    1% /mnt/e
tmpfs         762M    20K  762M    1% /run/user/1000
rahatut@rahatut-laptop:~/lab2$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.4Gi         639Mi        6.8Gi         4.5Mi         187Mi        6.8Gi
Swap:         2.0Gi           0B         2.0Gi
rahatut@rahatut-laptop:~/lab2$ top
top - 13:59:10 up  1:41,  1 user,  load average: 0.00, 0.00, 0.00
Tasks: 23 total,  1 running, 22 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.5 sy,  0.0 ni, 99.5 id,  0.0 wa,  0.0 hi,  0.0 s
MiB Mem :  7618.2 total,  6942.9 free,   642.0 used,   187.6 buff
MiB Swap: 2048.0 total, 2048.0 free,    0.0 used. 6976.2 avai

```

Figure 4: Execution Screenshot 4